Reciprocal cross

Whenever we cross a male possessing a specific trait with a female possessing a different trait, the procedure is called a *reciprocal cross*. Mendel created reciprocal crosses when he crossed female yellow peas with male green peas.

1 What would have been an alternative reciprocal cross using the same characteristics?

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Filial: to do with a son, daughter, child or offspring.

F_1 and F_2 generations

This is a shorthand term for 'first filial' and 'second filial' generations. The first generation progeny (offspring) of a reciprocal cross are called the F_1 generation. The abbreviation F_1 means, *'The resulting progeny of the reciprocal cross being studied'.*

2 Why do we usually say F_1 rather than use the complete definition given above?

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The F_2 generation is the progeny of crossing the F_1 individuals. Members of the F_1 generation can be allowed to self-fertilise (in plants) or cross-fertilise with other members of the F_1 generation. Either way, the progeny is referred to as the F_2 generation.

Cell cycle: growth and reproduction

All cells grow and reproduce. They use either of the processes mitosis or meiosis.

Mitosis occurs in *somatic* cells. These are cells used for growth and development of an organism.

Meiosis is exclusively used by *germ* cells. Germ cells are involved in reproduction. Male and female reproductive cells are referred to as *gametes*. When a female gamete (ovum) is fertilised by a male gamete (sperm) a *zygote* is formed.

Use terms mentioned previously on this sheet and match them to the following definitions/descriptions.

3 Takes place in the nucleus of a dividing cell and forms two new nuclei that have the same number of chromosomes as the parent nucleus. _____

4 A process causing the number of chromosomes in gamete-producing cells to be halved. _____

5 A mature male or female germ cell. _____

6 A type of cell that makes up tissues and organs in the body, apart from germ cells. _____

7 Results from the fusion of a male sex cell and a female sex cell. _____

8 Male gamete or germ cell. _____

9 Female gamete or germ cell. _____

10 Offspring of plants or animals. _____

11 Paired crosses where a male with a specific trait is crossed with a female possessing another trait, and vice versa.

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